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Research and Applications

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Social media use and mental health among older adults with multimorbidity: the role of self-care efficacy

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Abstract

Objectives: To describe the prevalence and trends in the use of social media over time and explore whether social media use is related to better self-care efficacy and thus related to better mental health among United States older adults with multimorbidity.

Materials and Methods: Respondents aged 65 years+ and having 2 or more chronic conditions from the 2017-2020 Health Information National Trends Survey were analyzed ($N = 3341$) using weighted descriptive and logistic regression analyses.

Results: Overall, 48% ($n = 1674$) of older adults with multimorbidity used social media and there was a linear trend in use over time, increasing from 41.1% in 2017 to 46.5% in 2018, and then further up to 51.7% in 2019, and 54.0% in 2020. Users were often younger, married/partnered, and non-Hispanic White with high education and income. Social media use was associated with better self-care efficacy that was further related to better mental health, indicating a significant mediation effect of self-care efficacy in the relationship between social media use and mental health.

Discussion: Although older adults with multimorbidity are a fast-growing population using social media for health, significant demographic disparities exist. While social media use is promising in improving self-care efficacy and thus mental health, relying on social media for the management of multimorbidity might be potentially harmful to those who are not only affected by multimorbidity but also socially disadvantaged (eg, non-White with lower education).

Conclusion: Great effort is needed to address the demographic disparity and ensure health equity when using social media for patient care.

Key words: social media; self-care efficacy; mental health; multimorbidity; older adults.

Introduction

Multimorbidity refers to the co-occurrence of 2 or more chronic conditions.[1] The prevalence rate of multimorbidity increases significantly with age, ranging from 15% to 43% among older adults worldwide, with a particularly higher rate of 65% among those aged 65 and older.[1–4] Compared to their counterparts without multimorbidity, individuals facing multiple chronic conditions encounter greater challenges in care management. These challenges involve managing complex symptoms, navigating conflicting advice, and adhering to different prescriptions, all of which can be distressing and demanding.[3] Self-care efficacy, defined as confidence in performing activities necessary to maintain and improve health,[5,6] is fundamental to managing a person's own chronic conditions and assuming primary responsibility for one's own health.[5–7] Self-care efficacy has been shown to contribute to favorable health outcomes, including mental health.[8,9] Most importantly, self-care efficacy can be learned from role models and improved by training and social support.[5,8] Nevertheless, research suggests that older adults with multimorbidity often struggle more with self-care and are more likely to lack self-care efficacy.[6,7] Recent evidence also highlights additional challenges, as older adults with multimorbidity are often excluded from clinical trials,[10] resulting in limited evidence-based guidance available for self-care in this population.

Social media tools, such as social networking sites like Facebook, LinkedIn, YouTube, and online forums and support groups, are traditionally considered a supplement to healthcare services.[11] However, with the increasing number of older adults utilizing the internet,[12] its potential to support self-care is gaining international interest.[13,14] Social media can help older adults engage in meaningful social activities and relationships and provide access to social support[11,15,16] that is likely to be lost in their networks as they age. Furthermore, social media might also enable older adults to access

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and share health information[17] and possibly improve self-care efficacy, including beliefs, knowledge, and skills to manage their condition,[12] through meaningful social support, social connections, and empowerment. All these benefits are invaluable for the health of older adults with multimorbidity given that they are more likely to feel isolated and struggle with loneliness and a sense of control over their self-care challenges.[18,19] Yet, simultaneously, concerns have also been raised that social media might exacerbate psychological distress and mental illness.[12] Studies on adolescents[20,21] and young adults[22] suggest that social media use might serve as a marker of underlying vulnerability and mental distress[15,22] and be associated with a high risk for depression.[22] Since much of the research on social media has focused exclusively on younger adults,[20–22] the investigation of older adults is relatively recent and limited. It is not clear whether the use of social media improves or impairs the mental health of older adults, particularly those with multimorbidity, or what the possible mechanisms may be by which social media influences mental health.

With the rise of new technology and increasing interest in using these technologies as digital solutions to support self-care,[13] it is important to understand whether older adults use social media tools and how they use them to manage their chronic conditions.[18] Furthermore, whether the use of social media is beneficial or detrimental might partially depend on self-care efficacy.[5] In this study, we aimed to (1) describe the prevalence of and trends in the prevalence of social media use over time, and (2) explore whether self-care efficacy serves as an important mediator in the relationship between social media use and mental health among older adults with multimorbidity. We hypothesized that social media use is related to better self-care efficacy and thus related to better mental health among United States older adults with multimorbidity. The positive effect of social media use is based on the evidence that self-care efficacy can be improved by positive use of social media for active involvement in self-care,[11,12] including sharing health information on social networking sites and participating in online support groups.

Methods

Data source

Data were extracted from multiple cycles of a cross-sectional survey of Health Information National Trends Survey 5 (HINTS 5)-Cycles 1-4 between 2017 and 2020. HINTS is a population-based, nationally representative survey of American adults with aims to understand how American adults access and use health information via information and communication technology.[23] HINTS 5 is a single-mode mail survey using a 2-stage complex design except for Cycle 3 which employed a web pilot alongside the mail survey. Detailed reports on the sampling design for the HINTS have been published elsewhere.[23] The data were publicly available and deidentified, neither consent nor institutional review board approval was required.

Combined data from HINTS 5-Cycles 1-4 resulted in a total of 16 092 respondents with 3285 from Cycle 1, 3504 from Cycle 2, 5438 from Cycle 3, and 3865 from Cycle 4. Respondents were included in the analysis if they were aged 65 years or older and had at least 2 chronic conditions including cancer, diabetes, hypertension, heart conditions, lung disease, arthritis, and depression except for Cycle 3 and Cycle 4 where arthritis was not assessed, yielding a final sample of 3341 (representing a population size of 1113 159 365) older adults with multimorbidity.

Measures

Social media use. Social media use was assessed using 1 question "In the past 12 months, have you used the internet for any of the following reasons?" with a binary response to the following categories: (1) to visit a social networking site, such as Facebook or LinkedIn, (2) to share health information on social networking sites, such as Facebook or Twitter, (3) to participate in an online forum or support group for people with a similar health or medical issue, and (4) to watch a health-related video on YouTube. Respondents who answered "yes" to any of these categories were classified as using social media and defined as social media users.

Self-care efficacy. Self-care efficacy was assessed by self-report on a single question asking "Overall, how confident are you about your ability to take good care of your health?" on a 5-point Likert scale from 1 = not confident at all to 5 = completely confident, with a high score indicating higher levels of confidence.

Mental health. Respondents were asked to rate how often they have been bothered by the following problems over the past 2 weeks: (1) little interest or pleasure in doing things, (2) feeling down, depressed, or hopeless, (3) feeling nervous, anxious, or on edge, and (4) not being able to stop or control worrying. A 4-point Likert scale ranging from 1 = nearly every day to 4 = not at all was used as a response scale. A summary score of these items was created to represent the levels of mental health, with a high score indicating better mental health. Cronbach's alpha in the current study is 0.87.

Covariates. Additional covariates included age (years), gender (0 = male, 1 = female), marital status (0 = separated, divorced, widowed, or single, 1 = married or partnered), education (0 = high school or less, 1 = some college or more), race/ethnicity (0 = Non-Hispanic White, 1 = Non-Hispanic Black or African American, 2 = Hispanic, 3 = other), annual household income (0 = less than US $50 000; 1 = US $50 000 or more), living alone (0 = living with others, 1 = living alone), rural residency (0 = nonrural, 1 = rural), and health care insurance coverage (1 = yes, 0 = no).

Data analysis

Weighted analyses using the jackknife approach were performed to provide nationally representative estimates with the HINTS-supplied final sample weights for population-level point estimates and 200 replicate weights for accurate standard errors. For *research question* 1, we first used descriptive statistics to describe the overall and yearly prevalence and compared the demographic characteristics of social media users and non-users. We then used logistic regressions with survey year as a predictor (reference year = 2017) adjusted for age and gender to test trends in the prevalence of social media use over time. To confirm the trend results, we also treated year as a continuous variable to test a linear trend and added a quadratic term of year to test a quadratic trend in the logistic regression models that adjusted for age and gender. The quadratic trend was not significant and was not reported in the results. For *research question* 2, we used mediation analyses to test whether social media use was associated with mental health via self-care efficacy. Two models were fit: Model 1 without covariates and Model 2 adjusted

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for covariates (eg, age, gender, marital status, and education). The indirect effect was tested using bootstrapping based on 2000 samples and was statistically significant if the 95% bias-corrected confidence intervals do not contain zero. We also performed a sensitivity analysis to explore the potential role of social media use as a mediator between self-care efficacy and mental health. In other words, self-care efficacy might have a global association with social media use, which in turn could influence mental health outcomes. All analyses were performed using Stata (Version 17 Stata Corp., College Station, TX, United States). A 2-sided $P < .05$ was statistically significant.

Results

Sample characteristics of social media users

Table 1 details the sample size and sample characteristics of social media users and non-users between 2017 and 2020. Among 3341 respondents (population size = 1113 159 365), the mean age was 74.89 years (standard deviation [SD] = 7.57, range = 65-104), and the mean number of chronic conditions was 2.80 (SD = 0.98, range = 2-7), with 49% reporting 2 conditions and 51% reporting 3 or more conditions. Most respondents were females (55%), married or partnered (54%), had some college or higher education (56%), and self-identified as non-Hispanic White (68%). Compared to social media non-users, users were more likely to be younger in age, married or partnered, non-Hispanic White, had some college or higher education, had a higher annual household income, with health care insurance coverage, and living with others. Furthermore, social media users reported higher levels of self-care efficacy, better mental health, and fewer chronic conditions.

The overall prevalence of and trends in social media use over time

Table 2 shows the prevalence of and changes in social media use over time among older adult respondents with multimorbidity. Overall, 48% ($n = 1674$) of respondents used social media and there was a linear trend in the prevalence of social media use over time (OR = 1.19, 95% CI = 1.09-1.30, $P < .001$), increasing from 41.1% in 2017 to 46.5% in 2018, and then increasing to 51.7% in 2019 and 54.0% in 2020. Adjusted linear trends for social media use remained significant (aOR = 1.22, 95% CI = 1.11-1.34, $P < .001$).

Test of mediation effects

Table 3 shows the results of the mediation analyses among older adult respondents with multimorbidity. Specifically, social media use was associated with better self-care efficacy ($\beta = 0.17$, 95% CI = 0.08-0.26, $P < .001$) and self-care efficacy was further related to better mental health ($\beta = 1.02$, 95% CI = 0.83-1.22, $P < .001$), indicating a significant mediation effect of self-care efficacy ($\beta = 0.11$, 95% CI = 0.05-0.18, $P = .001$) in the relationship between social media use and mental health. This mediation effect remained significant ($\beta = 0.10$, 95% CI = 0.03-0.17, $P = .005$) after controlling for age, gender, marital status, education, race/ethnicity, annual household income, living alone status, rural residency, healthcare insurance coverage, and the number of chronic conditions. The proportion of total association mediated through self-care efficacy was 29%.

Furthermore, we investigated if social media use served as a mediator between self-care efficacy and mental health through a sensitivity analysis. The results (Table S1) indicated that while there was a significant association between self-care efficacy and social media use, no significant relationship was found between social media use and mental health. This suggests that although self-care efficacy is positively linked to social media use, the effect of social media use on mental health was not clear.

Discussion

Principal findings

Social media has recently been seen as a digital solution to mental health issues not only for younger adults but also for older adults after the COVID-19 pandemic.[18] A primary driver for better mental health among older adults with multimorbidity is strong efficacy in taking care of one's own health.[5] Although much is known about the association between self-care efficacy and health outcomes (eg, mental health),[8,9] few studies have explored the association within the context of social media use, which is an emerging application of digital technology in the health field.[13,14] Using a representative sample of United States older adults with multimorbidity, we examined whether and how social media use affects mental health through self-care efficacy as well as user profiles. Several important findings are noted.

Overall, 47.95% of the United States older adults with multimorbidity indicated the use of social media. These prevalence estimates are similar to those reported in the general United States older adults who are free of multimorbidity (45%).[24] We also report a linear trend in social media use over time, with 41% in 2017, 47% in 2018, 52% in 2019, and 54% in 2020. This trend is similar to Chou's recent study,[15] which found an increasing trend in social media use across all demographic groups in America from 2007 to 2019. According to a recent report by the Pew Research Center,[24] older adults, despite being late in using social media, represent a fast-growing segment of the population who are potential users and benefit most from such technologies.[24,25] To our knowledge, we report the first population-based estimates of social media use in this population. This group of individuals with complex care needs are particularly vulnerable to mental illness and require lifelong self-care commitment. Our findings suggest that older adults are beginning to embrace social media in their lives,[25] supported by prior evidence indicating their desire to access health information through social media.[4,15] This implies that social media is promising in supporting the care needs of this vulnerable population. Additionally, broadly consistent with previous studies on different types of technology and populations, we found that social media use is more common in those at younger ages,[15,25–27] and non-Hispanic Whites,[27,28] and is increased by levels of education and income.[15,25,28] For example, in a study of general digital technology (including social media), less than 20% of older adults aged 65 years+ reported the use of these technologies, with those who are older and Black and Latino being less likely and those who are higher in income and education (eg, college education) being more likely to use technologies.[28] Our findings together with previously reported findings suggest that demographic disparities exist in the use of digital technology in general and social media in particular for older adults. Such a disparity in

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Table 1. Sample characteristics of social media users and non-users, 2017-2020.

Characteristics	All ($N = 3341$, population size = 113 159 365)	Nonusers ($n = 1667$, population size = 58 895 986)	Users ($n = 1674$, population size = 54 263 379)	*P* value
Age (years), mean (SD weighted)	74.89 (7.57)	76.93 (7.76)	72.68 (6.61)	*<.001*[a]
Self-care efficacy, mean (SD weighted)	3.75 (0.88)	3.67 (0.91)	3.83 (0.84)	*<.001*
Mental health, mean (SD weighted)	13.96 (2.98)	13.75 (3.13)	14.19 (2.77)	*.006*
Number of comorbidities, mean (SD weighted)	2.80 (0.98)	2.85 (0.99)	2.74 (0.96)	*.030*
Female (% weighted)	54.6	54.2	56.7	.310
Married or partnered (% weighted)	54.0	48.0	62.3	*<.001*
Some college or more (% weighted)	56.2	47.1	67.3	*<.001*
Race/ethnicity (% weighted)				*.002*
Non-Hispanic White	67.6	75.2	82.6	
Non-Hispanic Black or African American	7.6	10.2	7.5	
Hispanic	7.1	10.6	6.1	
Other	3.3	4.0	3.8	
Annual household income (≥ $50 000) (% weighted)	33.6	29.0	50.3	*<.001*
Living alone (% weighted)	30.9	37.1	26.1	*.001*
Rural residency (% weighted)	17.0	18.1	15.8	.210
Having health care insurance coverage (% weighted)	97.6	98.9	99.7	*.048*

Codes for demographic variables were age (years), gender (0 = male, 1 = female), marital status (0 = separated, divorced, widowed, or single, 1 = married or partnered), education (0 = high school or less, 1 = some college or more), race/ethnicity (0 = Non-Hispanic White, 1 = Non-Hispanic Black or African American, 2 = Hispanic, 3 = other), annual household income (0 = less than US $50 000; 1 = US $50 000 or more), living alone (0 = living with others, 1 = living alone), rural residency (0 = nonrural, 1 = rural), and health care insurance coverage (1 = yes, 0 = no).

[a] Italicized values are significant at $P < .05$.

Table 2. Prevalence of and trends in social media use over time, 2017-2020 ($N = 3341$, population size = 113 159 365).

	Prevalence		Trends					
	Percent	95% CI	OR[a]	95% CI	*P*	aOR[b]	95% CI	*P* value
2017	41.09	36.83, 45.50	Ref			Ref		
2018	46.51	41.78, 51.32	1.24	0.96, 1.62	.10	1.25	0.94, 1.67	.130
2019	51.72	46.73, 56.60	1.54	1.17, 2.01	*.002*[d]	1.59	1.20, 2.11	*.001*
2020	53.95	48.78, 59.04	1.68	1.33, 2.37	*<.001*	1.77	1.33, 2.37	*<.001*
Overall	47.95	45.56, 50.36	–	–	–	–	–	–
Test for a linear trend[c]	–	–	1.19	1.09, 1.30	*<.001*	1.22	1.11, 1.34	*<.001*

Data are weighted. OR = odds ratio; CI = confidence interval.

[a] OR = unadjusted odds ratio.

[b] aOR = adjusted odds ratio that accounted for age and gender.

[c] Logistic regression models adjusted for age and gender were used to conduct a test for linear trends. Year was entered as a continuous variable in the analysis; a quadratic term for year was also included but found to be insignificant and therefore not reported.

[d] Italicized values are significant at $P < .05$.

older adults may have been accelerated during the COVID-19 pandemic. Significant disparities emphasize the need to address this digital divide in older adults, particularly those who are not only affected by multiple chronic conditions but also socially disadvantaged (eg, Hispanic or Black with lower education) and those may be experiencing visual or cognitive limitations that impede meaningful use of such tools.

Many studies have identified a paradoxical effect of social media use and mental health[12,28,29] with some indicating a beneficial effect, whereas others indicating a detrimental effect. To resolve the paradoxical effect of social media use on mental health in older adults with multimorbidity, we analyzed the data based on the key assumption that whether social media use is beneficial or detrimental might partially depend on self-care efficacy.[8,18] Consistent with our hypothesis, we found that social media use exerts its impact on mental health through self-care efficacy. Specifically, older adults who used social media reported great confidence in performing self-care activities and thus reported better mental health. Two main reasons can explain this finding. First, our analysis showed that social media use is more likely to be used for health purposes and supportive in nature. Examples of such supportive activities include using social media to share health information, reaching out for online support, and watching health-related videos (whereas visiting networking sites more broadly might not necessarily be meaningful for users). Social media might serve as a useful tool for users to connect to individuals with similar health conditions and concerns and develop health-related identities that may build mutual support and shared trust.[13,14] Positive online support can facilitate the sharing of disease diagnosis, symptom management, and treatment planning.[14] Social media for such applications is very likely to be positive and promising to promote positive outcomes (eg, reducing social isolation and overcoming loneliness and depressive symptoms during the COVID-19 pandemic).[11–13] Additionally, prior research indicated that, unlike young adults, older adults who use social media are less likely to use it for entertainment and self-expression, but rather for

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Table 3. Mediation results ($N = 3322$, population size = 112 601 475).

	Model 1			Model 2		
	ß	95% CI	*P*	*ß*	95% CI	*P* value
Self-care efficacy						
Social media use	0.17	0.08, 0.26	*<.001*[a]	0.17	0.08, 0.26	*<.001*
Mental health						
Self-care efficacy	1.02	0.83, 1.22	*<.001*	1.00	0.78, 1.22	*<.001*
Social media use	0.27	−0.03, 0.57	.080	0.08	−0.25, 0.41	.650
Indirect effect	0.11	0.05, 0.18	*.001*	0.10	0.03, 0.17	*.005*

CI = confidence interval. Data are weighted. Model 1 was unadjusted estimates; Model 2 adjusted for age (in years), gender (0 = male, 1 = female), marital status (0 = separated, divorced, widowed, or single, 1 = married or partnered), education (0 = high school or less, 1 = some college or more), race/ethnicity (0 = Non-Hispanic White, 1 = Non-Hispanic Black or African American, 2 = Hispanic, 3 = other), annual household income (0 = less than US $50 000; 1 = US $50 000 or more), living alone (0 = living with others, 1 = living alone), rural residency (0 = nonrural, 1 = rural), health care insurance coverage (1 = yes, 0 = no), and the number of chronic conditions.

[a] Italicized values are significant at $P < .05$.

meaningful relationships (eg, stay connected with their families)[19] and better health both mentally and cognitively.[15,25,30] These results indicate older adults might be more likely than young adults to benefit from social media use. Future research is needed to use longitudinal data to compare the purposes of social media use between younger and older adult populations and explore how different purposes of social media use might be associated with mental health across different populations.

Alternatively, self-efficacy in self-care is an important protective factor for mental health among older adults with co-occurrence of multiple conditions. Although self-care efficacy has not been previously explored in our population within the context of social media use, self-efficacy is generally considered a strong indicator of inner strength and a sense of mastery that empowers individuals to better adapt to psychological distress and manage their illnesses.[6,7] Importantly, self-efficacy can be improved through personal mastery experiences, social support, and role modeling which has been the central target of many patient training and intervention programs.[8] Managing multiple conditions is a complex process and this process can be further complicated by the lack of guidance on disease management[3] as researchers tend to consider chronic conditions as separate rather than coexisting conditions.[31,32] Unmet care needs and unmanaged conditions can dramatically exacerbate existing psychological distress.[7,33] Our analysis suggests that social media use (possibly for health purposes) is positively related to self-care efficacy, suggesting self-care efficacy can be improved by digital platforms, like social media, and self-care efficacy also possibly helps to meet the unmet needs in this population. This finding is in line with prior evidence that self-efficacy can be improved by various sources,[6] including social media, and can help identify optimal options, prioritize competing needs, and optimize treatment decisions,[6] which is the key to successful chronic condition management. All the findings further suggest that social media could serve as a new platform for health interventions and educational programs to reach populations who are typically excluded from clinical trials, such as older adults with multimorbidity,[10] and to deliver efficacy-based interventions for better self-care activities when implemented with an inclusive design focus.[19]

Limitations

This study has several limitations. First, from the cross-sectional data, we cannot disentangle whether mental distress leads to social media use or vice versa, given that older adults with multimorbidity are very likely to be depressed. Similarly, we are not able to draw a causality but rather take advantage of population-level data to explore the possible mechanisms by which using social media might be beneficial for mental health. Second, self-reported measures might be vulnerable to recall bias and social desirability, future work using objective measures such as wearable devices or smartphone activity trackers for social media use behaviors is needed to confirm the current study. Thirdly, the trend in social media use in our analyses is based on the cross-sectional data, future work using longitudinal data to track individual patterns of change is needed to confirm the current finding. Our study exclusively focused on older adults with multimorbidity. We did not conduct in-depth analysis on other comparison populations, such as older adults without multimorbidity or younger adults with multimorbidity. Consequently, we are unable to draw distinct conclusions regarding whether social media use is associated with better mental health through improved self-care efficacy solely for the study population. Future research should delve into the underlying mechanisms of social media use and mental health in different populations.

Conclusion

While the use of social media is increasing in older adults with multimorbidity, it is not reaching all older adults, particularly those who are socially disadvantaged. Significant disparities exist in the use of social media among this particularly vulnerable population. Although social media use was associated with self-care efficacy and subsequently better mental health, this association in tandem with significant use disparities is most concerning for healthcare providers who deliver care and provide intervention via social media. Future research using longitudinal data with more accurate measurement is needed to determine the relationships between sociodemographics and social media use, and relationships between social media use and health outcomes including self-care efficacy among older adults with multimorbidity.

Author contributions

Zuoting Nie, Shiying Gao, Long Chen, Linda S. Edelman, Katherine A. Sward, Yun Jiang, and George Demiris (data interpretation and manuscript writing); Rumei Yang (study

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design, data acquisition, analysis, and interpretation, and preparation of the manuscript).

Supplementary material

Supplementary material is available at *Journal of the American Medical Informatics Association* online.

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Conflicts of interest

None declared.

Data availability

The data underlying this article are public and can be available in Health Information National Trends Survey at https://hints.cancer.gov/.

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